

PART I: FUNCTIONAL SPECIFICATION & SYSTEM ARCHITECTURE

1. Project Charter & Multi-Tier Global Admin Hierarchy

The platform is operated under a strict three-tier administrative hierarchy. This structure is designed to isolate regional operations while automating revenue distribution and compliance audits.

1.1 Commission Split and Payout Logic

The Clox platform charges a **30% gross platform fee** on all transactions, leaving a **70% net payout** for the Transportation Company executing the trip. The 30% platform fee is divided as follows:

- **Super Admin (Clox HQ - Global):** Retains **15% of the gross fare** to cover global server infrastructure, security, international payment processing, and 3rd-party API costs.
- **Regional Master Admin (State/Province):** Receives **10% of the gross fare** for loads originating in their designated state/province. Responsible for regional audits.
- **Local BDE Admin (City/Suburb):** Receives **5% of the gross fare** for loads originating in their local suburb/city.

1.2 The Fortnightly (4th Night) Admin Payout Sequence

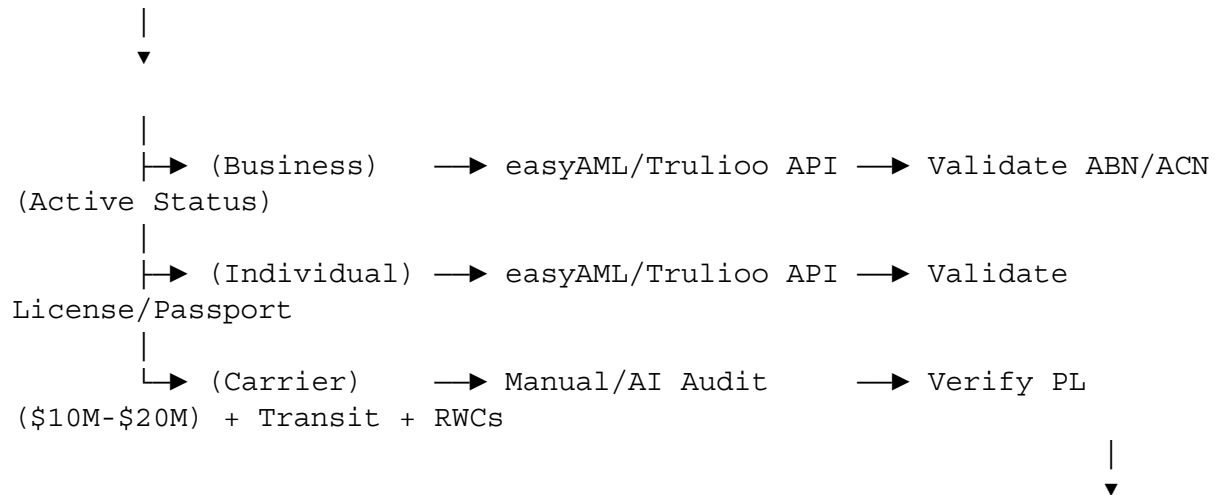
Admin commissions (State and Local BDE) are calculated over a rolling 14-day cycle and distributed every **4th night** (fortnightly). Before the transfers are processed, the system automatically subtracts **Management Fees** and **Marketing Costs** from the admin's gross share using the following formula :

$$\text{Net Admin Payout} = \text{Gross Admin Share} - \text{Management Fees} - \text{Marketing Costs}$$

All admin payouts are executed via **Monoova (NPP/Osko)** for real-time bank account settlement.

2. Multi-Tenant Identity & Compliance Gate

Cloxx implements a strict multi-tenant data isolation architecture to support worldwide deployment while keeping regional data legally compliant (e.g., GDPR in Europe, Privacy Act in Australia).



2.1 Sender Onboarding Paths

The system splits onboarding into two distinct validation tunnels:

- **Business Senders:** Must provide an Australian Business Number (ABN) or Australian Company Number (ACN). The system uses **easyAML** to check the Australian Business Register (ABR) in real time to ensure the business status is active.
- **Individual Senders:** Must provide a full residential address and scan a valid government-issued photo ID (Driver's License or Passport). easyAML cross-references this against the Document Verification Service (DVS).

2.2 Transportation Company "Compliance Gate"

A carrier cannot view, bid on, or accept jobs until their profile has been unlocked by a Regional Admin. During onboarding, carriers must upload:

1. **ABN/ACN** (real-time status verified).
2. **Public Liability Insurance Certificate** (with a mandatory minimum value of \$10,000,000 to \$20,000,000).
3. **Transit/Cargo Insurance Certificate**.
4. **Roadworthy Certificates (RWC)** for each vehicle, with the system pulling the expiry date into the maintenance watchdog.

3. Sender's RFP & Multi-Pickup Routing Workflow

Senders do not book a specific truck or driver; instead, they create a Request for Proposal (RFP) that is broadcast to qualified carriers.

3.1 Job Selection Modes

- **Interstate / Regional (Per KM):** Fixed pricing range with a single pickup and single drop-off point.
- **Local (Hourly):** Dedicated vehicle and driver booked on a "Day Basis" with a **mandatory minimum charge of 4 hours**, even if the job takes less time, to remain compliant with the *Road Transport and Distribution Award 2020*.

3.2 Hourly Multi-Pickup & Multi-Drop Logic

To allow flexible city distribution, hourly jobs support multi-stop routing :

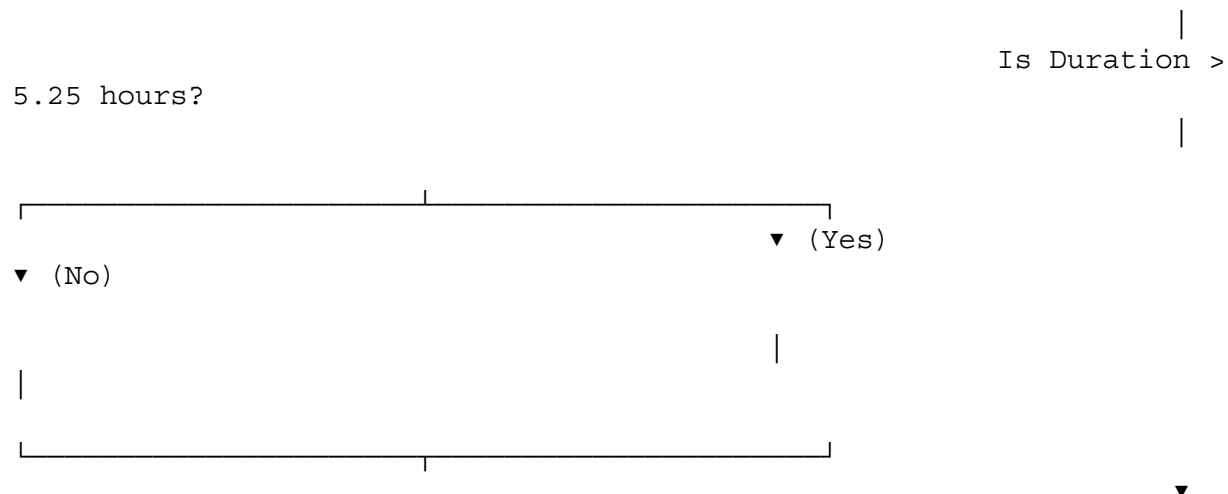
- **Pattern A (Multi-Pickup):** Maximum of 3 Pickups → 1 Final Drop-off.
- **Pattern B (Multi-Drop):** 1 Pickup → Maximum of 3 Separate Drop-offs.

3.3 Auto-Route Planning & Driver Fatigue Gate

For multi-stop bookings, the system auto-sequences the locations using a basic Traveling Salesman Problem (TSP) algorithm via **Google Routes API** to optimize travel time and distance.

The system must check the total planned duration (including loading/unloading window offsets). If the estimated duration exceeds **5.25 hours**, the system must automatically insert an **unpaid 15-minute rest break** into the ETA timeline displayed to the sender, ensuring compliance with the National Heavy Vehicle Regulator (NHVR).

→ →



3.4 Load Specifications & Smart Safety Rules

Senders must enter the physical weight (dead weight in kg) and dimensions (L x W x H). The system uses checkboxes for load types and applies auto-logic rules:

Load Type	Mandatory Checkbox Sub-Options	Hardcoded System Safety Rule
Dangerous Goods (DG)	Class selection (1-9) & Weight input	Only broadcast to carriers with verified DG-licensed drivers and trucks with proper placards. Mandatory manifest document upload if weight > 1,000\text{ kg}.
Oversize / Over-dimension	Length > 13.6\text{ m} or Width > 2.5\text{ m}	Triggers permit warning; limits bids to flatbed/tray vehicles only.
Liquid / Bulk Liquid	Drums, IBCs, or Tanker	Requires confirmation of spill containment equipment on the truck.
Steel / Metal	Coils, Rebar, or Plates	Driver must confirm carriage of heavy-duty chains and NHVR-compliant cargo restraints.
Temperature Controlled	Frozen, Chilled, or Pharma	Requires refrigerated vehicle; temperature range entry is mandatory.
Containers	20ft or 40ft	Verifies that the assigned carrier has active port/wharf time slots and access permits.

3.5 Site Access & Maneuverability Assessment (Mandatory Wording)

To prevent futile trips and driver delays, Senders must complete a mandatory site assessment before publishing the RFP :

- **Maneuverability Profile:** Senders must select: Tight Street, Standard Industrial, or Wide Yard.
- **Facility Specs:** Senders must select: Raised Loading Dock (48-52 inches), Ground Level, or Forklift Required.
- **Legal Protection Disclaimer:** Senders must check a box agreeing to: *"Incorrect load type or site access selection may result in job rejection, futile travel fees, or additional waiting charges."*

4. Carrier Workflow: Basic Marketplace vs. Fleet+ SaaS

Clox provides two distinct modes for transportation companies to manage their business.

4.1 Basic Mode (Free Marketplace Access)

Provides core transactional brokerage capabilities:

- **Net Payout Display:** The job board and bidding interface display only the **Net Payout (70%)** that the carrier will receive after commission, not the gross sender price.
- **Asset Allocation Gate:** A carrier cannot submit a proposal unless they assign a specific, verified truck and driver from their registered fleet to that specific time slot.
- **Smart Conflict Handling:** If a carrier bids a specific truck on multiple overlapping jobs, the moment one job is awarded, all other bids for that vehicle automatically expire, and the carrier is notified instantly to prevent double-booking.

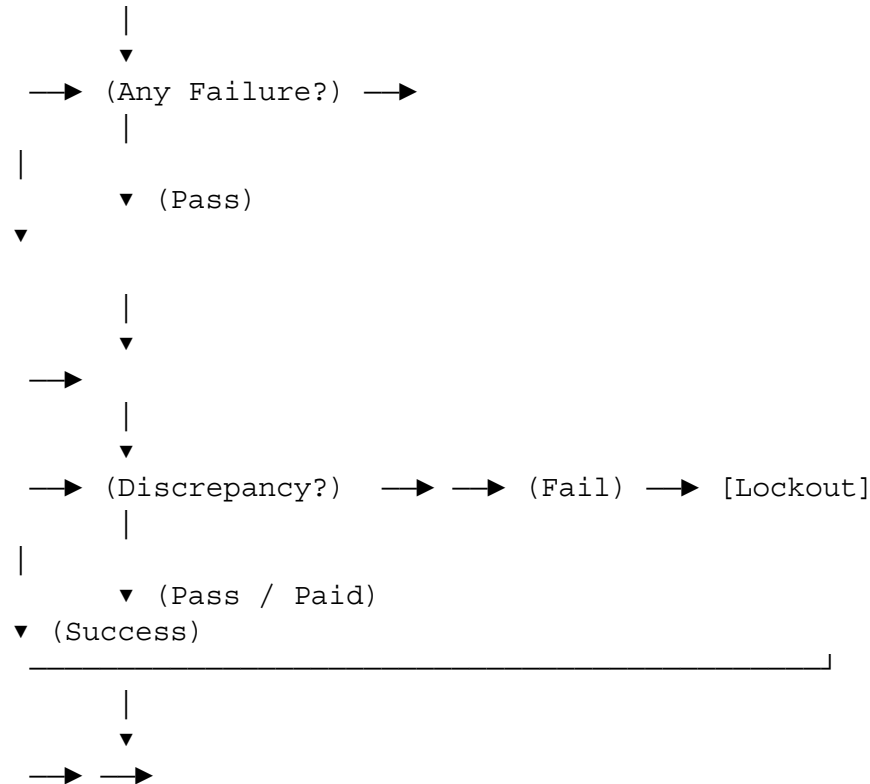
4.2 Fleet Management Mode (Fleet+ SaaS Subscription)

An optional, subscription-based tier (Starter, Growth, or Enterprise) that acts as an in-house operating system :

- **Cost & Profit Engine:** Automates trip-level profitability calculation using real-time net payouts and variable operating costs:
$$\text{Profit}_{\text{Trip}} = \text{Net Payout (70\%)} - \left(\text{Fuel Cost} + \text{Driver Wages} + \text{Tolls} + \left(\text{Maintenance Rate} \times \text{Distance} \right) \right)$$
- **Asset Depreciation (Enterprise Tier Only):** Subtracts a daily asset depreciation value based on the vehicle's initial purchase price and target residual value.
- **Operational Watchdogs:** Flags any vehicle on the dashboard if its profit margin drops below 15% or if its idle time exceeds 40%.
- **Maintenance Watchdog Lockout:** A system service that checks odometer readings against service intervals, and RWC expiry dates. If a vehicle exceeds these limits, the system auto-statuses the vehicle to MAINTENANCE and blocks it from job bidding.

5. Driver Workflow: The Safety-First Sequence

The Driver App is engineered with hardcoded legal gates to protect Achieve Global Enterprises from Chain of Responsibility (CoR) safety liability.



5.1 Gate A: Pre-Trip Safety Walk-around

Before commencing a shift, the driver must complete a physical checklist within the app. If any safety critical item (e.g., tyres under 1.5mm tread, soft brakes, or fluid leaks) is marked "Fail," the vehicle status changes to LOCKED in the database, a maintenance work order is created for the Transport Company, and the driver cannot start any trips.

5.2 Maneuverability Preview

Upon passing the checklist, the driver is shown a "Site Access Preview" displaying the Sender's turning circle and dock specs so they can plan their approach safely.

5.3 Geofence Arrival & Wait Time

When the truck enters the 200m geofence created by **Radar.com**, the system records the exact arrival timestamp.

- **At Pickup:** The system starts a 30-minute free loading window. After 30 minutes, wait-time billing begins, and push notifications are sent to all users.
- **At Destination:** Senders receive a 15-minute pre-arrival alert. Once arrived, a 60-minute free unloading window begins before wait-time charges accrue.

5.4 Gate B: Pickup Mass Verification (Discrepancy Flow)

Before loading, the driver must inspect the freight and secure it with appropriate straps or chains. If the driver identifies that the cargo is heavier than declared, they must tap **"Report Discrepancy"** and upload actual weight details and photo evidence (e.g., weighbridge ticket or cargo labels) :

1. The system calculates the surcharge based on the extra weight.
2. The Sender is notified: *"Additional charge of \$X required to proceed due to weight discrepancy."*
3. The system initiates a **Stripe Incremental Authorization** to capture the funds immediately.
4. **The Master Lock:** The **"Start Trip"** button is disabled. It only unlocks once the incremental authorization is successfully processed, or the sender pays the difference.

5.5 Fatigue Management (Manual Logbook)

In this phase, drivers manage fatigue and rest breaks manually using their physical national paper diaries. No automated in-app fatigue alerts are triggered. However, the driver has a "Taking Break" button; tapping this pauses the live GPS stream and updates the ETA for the sender.

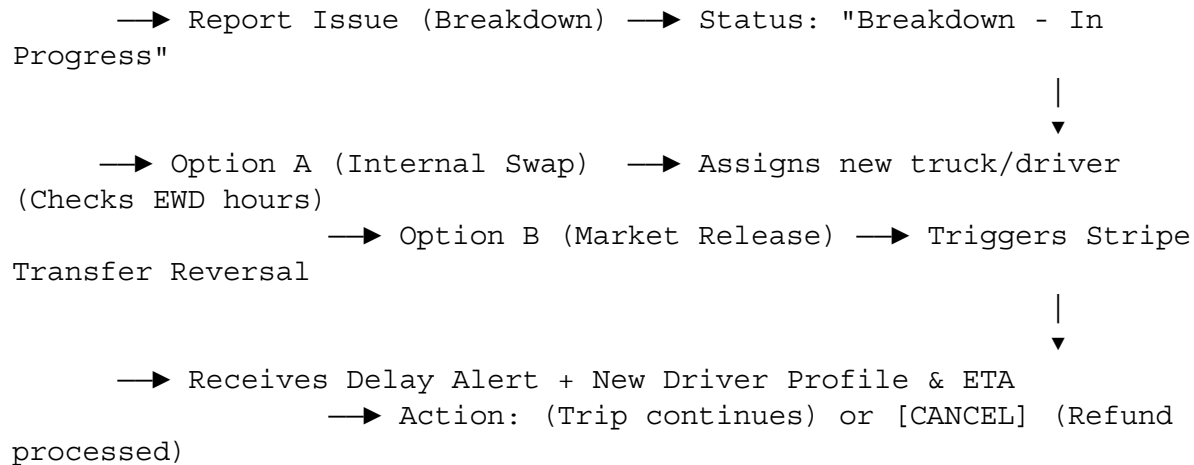
5.6 Proof of Delivery (POD) Capture

When the truck enters the destination geofence, the system locks the arrival timestamp. To finalize the trip, the driver must:

- Collect the receiver's **Sign-on-Glass (SOG) signature**.
- Enter the final receiver's legal name.
- Capture a mandatory photo of the offloaded goods at the site.
- The system bundles these into a PDF with a **server-locked, tamper-proof timestamp**.

6. Truck Breakdown & Exception Recovery Flow

The breakdown sequence is managed across all three user applications and the administrative dashboard to handle asset recovery and payment routing.



1. **Driver App:** Driver taps  **Report Issue** → selects **Truck Breakdown** → enters odometer and uploads a mandatory GPS and photo of the issue. Trip status is set to  Breakdown – In Progress.
2. **Transport Company Dashboard:** Receives an instant notification. The operator has three options:
 - **Option A (Internal Replacement):** Assigns a different truck and driver from their own registered fleet. The fare remains with the company.
 - **Option B (Release to Marketplace):** The job is cancelled, and returned to the open broadcast. The platform automatically triggers a **Stripe Transfer Reversal** to claw back the 70% net fare from Carrier A's pending balance to allow Carrier B to be paid.
 - **Option C (Repair & Continue):** The operator inputs a repair ETA, which must be within the driver's legal working hours.
3. **Sender App:** Sender receives a push notification explaining the delay. If a replacement is assigned, the app displays the new driver's photo, rating, vehicle registration, and updated ETA. Senders must tap **Approve** (to resume the trip) or **Cancel** (refunds are governed by the Australian Consumer Law major failure thresholds).
4. **Admin Panel:** Super Admin monitors the breakdown. The incident is automatically logged against the carrier's **Company Performance Score**.

7. Technical Infrastructure & 3rd Party Costs (AUD)

To minimize the technical "Connection Gap" and maintain global scalability, the core application relies on the following standardized third-party services :

- **Stripe Connect (Payment Orchestration):** Manages the 100% upfront escrow hold and split payouts.
 - *Cost:* \$2.00 AUD per active connected account/month + 2.9% + 30c per credit card transaction.
- **easyAML / Trulioo (Identity & KYB):** Automates ABN verification for businesses and passport/driver license verification for individuals.
 - *Cost:* \$1.00 AUD to \$3.00 AUD per check.
- **Radar.com (Geofencing & ETA):** Tracks isochrone-based travel-time alerts and arrival timestamps.
 - *Cost:* Pro tier starts at approximately \$600 AUD/month (supports up to 10,000 active users).
- **Twilio Proxy (Participant Number Masking):** Assigns temporary numbers to bridge calls and SMS between driver and sender without exposing personal data.
 - *Cost:* Approximately \$1.15 AUD per local phone number/month + usage (\$0.01 per SMS/minute).
- **Monoova (NPP/Osko Payouts):** Handles instant payouts to regional admins and carriers via the Australian New Payments Platform (NPP).
 - *Cost:* Flat fee of \$0.50 AUD to \$1.50 AUD per payout transaction.
- **Google Maps Platform (Routing & Maps):** Powers address completion and route calculation.
 - *Cost:* Usage-based per request (after standard \$200 free monthly credit).

8. Master Database ERD & Fleet API Schema

The database model is multi-tenant and branch-isolated to separate local transactions while allowing regional admin roll-ups.

8.1 Primary Entity Relationships (ERD Mapping)

- **Users Table:** user_id (PK), email, phone, role (Sender, Driver, Carrier, Admin).
- **Senders Table:** sender_id (PK), user_id (FK), sender_type (Business, Individual), abn_acn (nullable), verified_status.
- **Carriers Table:** carrier_id (PK), user_id (FK), abn, public_liability_status, transit_insurance_status, fleet_tier (Basic, Fleet+).
- **Vehicles Table:** vehicle_id (PK), carrier_id (FK), rego_number, gvm_gcm, rwc_expiry, status (Available, Active, Maintenance, Locked).
- **Trips Table:** trip_id (PK), sender_id (FK), carrier_id (FK), vehicle_id (FK), driver_id (FK), origin_suburb, destination_suburb, status (Created, Arrived_Pickup, Mass_Verified, In_Transit, Complete), gross_fare, net_carrier_fare (70%).
- **Admins Table:** admin_id (PK), user_id (FK), admin_type (Super, Master, Local), allocated_suburb_or_state.

8.2 Upgraded Fleet Registration API Specification

Used to onboard carrier vehicles while capturing the custom parameters needed for the Fleet+ subscription analytics.

- **Endpoint:** POST /v1/fleet/register
- **Header:** Authorization: Bearer <JWT>

JSON Request Body Payload:

```
{
  "registration_number": "REG-882",
  "vin_number": "1FTFW1EF5GXXXXXX",
  "vehicle_class": "3T_RIGID",
  "tare_weight_kg": 2800,
  "gvm_gcm_kg": 5800,
  "rwc_expiry_date": "2027-04-15",
  "insurance_policy_id": "PL-INS-99218",
  "cost_per_km": 1.25,
  "fuel_type": "DIESEL",
  "service_interval_km": 15000,
  "last_service_odo": 45000
}
```

JSON Response Payload (Success):

```
{
  "status": "SUCCESS",
  "message": "Vehicle registered successfully under Basic profile.",
  "data": {
    "vehicle_id": "veh_clox_88291A",
    "registration_number": "REG-882",
    "gvm_gcm_kg": 5800,
    "rwc_expiry_date": "2027-04-15",
    "maintenance_watchdog_status": "COMPLIANT",
    "analytics_tier": "BASIC_LIMITED"
  }
}
```

9. Clox Official Run Sheet & PDF Invoicing Specification

For local hourly bookings, the platform must automatically generate a downloadable PDF Run Sheet upon trip completion. This acts as the legal record of truth for billing audits, carrier payroll, and tax.

9.1 The 4-Hour Billing Floor Logic

For hourly jobs, the billing engine must execute a validation check upon POD :

$\text{Actual Duration} = \text{Final Drop-off Departure Time} - \text{Stop 1 Geofence Arrival Time}$

- If $\text{Actual Duration} < 4.0 \text{ hours}$, the invoice and run sheet must display **"Billed Duration: 4.0 Hours (Minimum Engagement)"**.

9.2 ATO Invoicing Standard

To comply with Australian Taxation Office (ATO) rules, if the final gross invoice value (including wait times and surcharges) is **equal to or greater than \$1,000 AUD**, the auto-generated PDF invoice **MUST** include the Sender's legal name and residential or business address. If under \$1,000, these details are optional but recommended.

9.3 Odometer Discrepancy Flag

The system must compare the GPS-tracked distance against the manual odometer entries input by the driver at the start and end of the shift. If:

$\text{Manual Odometer Distance} - \text{GPS Distance} > 10\%$

The run sheet is marked with a **"Odometer Discrepancy Flag"** and routed to the Super Admin dashboard to prevent billing fraud.

9.4 Clox Official Run Sheet Layout (PDF Template)

```
=====
=====
                                CLOX OFFICIAL RUN SHEET (LOCAL HOURLY)
=====
=====
Job ID: CLX-992182                                Date: 13 July 2026
Driver: John Doe                                  Carrier: Melbourne
Transport Pty Ltd
Vehicle Rego: REG-882                            Carrier ABN: 12 345 678
910
=====
=====
STOP TIMELINE & ODOMETER LOG:
-----
-----
Stop # | Type      | Address                               | Arrived | Departed |
Wait Time | Odo
-----
-----
1      | Pickup  | 12 Warehouse St, VIC                | 08:00 AM | 08:45 AM | 15
mins   |         |                                     |          |          |
2      | Pickup  | 45 Distribution Rd, VIC              | 09:15 AM | 09:40 AM | 10
mins   | (GPS)   |                                     |          |          |
3      | Drop    | 88 Retail Blvd, VIC                 | 10:20 AM | 11:15 AM | 25
mins   | (GPS)   |                                     |          |          |
4      | Drop    | 10 Final Way, VIC                   | 11:45 AM | 12:30 PM | 15
mins   |         |                                     |          |          |
-----
-----
TRIP SUMMARY & FATIGUE LOG:
-----
-----
* Total Duration: 4 Hours 30 Mins                * Billed Duration: 4.5
Hours
* Total GPS Distance: 60.5 km                     * Manual Odo Distance:
65.0 km
* Odometer Status: VERIFIED (<10% variance)

* Scheduled Breaks: None (Duration <= 5.25 hours)
* Driver Rest Signature:

* Final Receiver: Mehulkumar Patel                * Receiver Signature:
=====
=====
```

This document was generated by Clox.

PART II: CONTRACTS & LEGAL PROTECTION

1. Non-Disclosure Agreement (NDA)

BETWEEN: ACHIEVE GLOBAL ENTERPRISES PTY LTD (ABN 48 626 269 387) of Victoria, Australia, represented by its Director, Mehulkumar Patel (hereinafter referred to as the **"Disclosing Party"**).

AND: **** (Registration / ABN: _____) of [Insert Address], represented by its authorized representative, (hereinafter referred to as the **"Recipient"**).

1.1 Purpose

The Disclosing Party is sharing proprietary information regarding the "Clox" global freight forwarding marketplace to enable the Recipient to submit software development bids and build the platform.

1.2 Definition of Confidential Information

"Confidential Information" includes all financial, technical, and operational data, including:

- The Clox proprietary hardcoded safety sequence (SafetyCheck \rightarrow MassCheck \rightarrow TripStart).
- The multi-tier admin commission split (15% HQ / 10% State / 5% Suburb) and the automated deduction engine.
- The "Fleet+" cost-profit analytics and maintenance watchdog logic.
- The source code, database ERD schemas, and API configurations.

1.3 Confidentiality Obligations

The Recipient agrees to:

- Keep the Confidential Information strictly confidential and not disclose it to any third party without written consent.
- Limit internal access only to employees or subcontractors who "need to know" and are bound by identical confidentiality agreements.
- Not copy, reverse-engineer, or use the information to build a competing logistics platform.

1.4 Term & Survival

This Agreement is effective upon signature and survives for a period of **five (5) years** from the date of last disclosure, regardless of whether a formal development contract is executed.

1.5 Governing Law

This Agreement is governed by the laws of **Victoria, Australia**, and the parties submit to the exclusive jurisdiction of the courts of Victoria.

Signed by Disclosing Party (Mehulkumar Patel, Director): _____ **Date:**

Signed by Recipient (Developer Representative): _____ **Date:**

2. Software Development & Milestone Agreement

PARTIES:

1. **ACHIEVE GLOBAL ENTERPRISES PTY LTD** (ABN 48 626 269 387) of Victoria, Australia ("the Client").
2. **** (ABN / Rego: _____) ("the Developer").

2.1 Scope of Work

The Developer agrees to build, test, and deploy the "Clox" digital freight marketplace in strict compliance with the **Functional Specification Document (v25)**.

2.2 Payment Milestones & Deliverables

The total contract value is ** AUD ex-GST**, payable in five stages. **No payment shall be released without the completion of the respective milestone deliverables and written approval from Mehulkumar Patel (Director).**

Stage	Payment %	Milestone Deliverable Requirements	Trigger for Release
Stage 1	10%	System Architecture & Clickable Wireframes: Delivery of FSD, ERD Database Model, API Specifications, and low-fidelity clickable mockups of all user screens.	Written approval of design & database structure by Mehulkumar Patel.
Stage 2	20%	Frontend UI Build: Completed frontend screens for Sender App, Transport Co. Dashboard, Driver App, and Admin Panel.	Frontend UI delivered to staging environment for manual interaction.
Stage 3	30%	Backend Core & Escrow Integration: Completed RFP bidding engine, Stripe Connect 100% upfront escrow setup, and Monoova NPP admin/carrier split payouts.	Integration tested successfully on Stripe/Monoova sandboxes.
Stage 4	20%	Safety Sequence & Geofencing: Integration of Radar.com	Successful simulation of safety sequence from checklist to POD.

Stage	Payment %	Milestone Deliverable Requirements	Trigger for Release
		(Geofencing), hardcoded driver Pre-Trip Checklist, Mass Check Discrepancy flow, and live tracking map.	
Stage 5	20%	UAT, Handover & Launch: Final User Acceptance Testing (UAT), bug fixes, transfer of all IP, server deployment, and delivery of system manuals.	Production launch on iOS/Android Stores and complete code handover.

2.3 Intellectual Property (IP) Ownership

The Developer agrees that **all source code, database structures, designs, graphics, and trade secrets** produced under this Agreement are the sole and exclusive property of **Achieve Global Enterprises Pty Ltd** upon payment of each respective milestone. The Developer shall sign an IP Assignment Deed at Stage 5.

2.4 Regulatory Warranty

The Developer warrants that the software will be built to comply with:

- **NHVR (AU):** Chain of Responsibility (CoR) mass and dimension limits.
- **Fair Work (AU):** 4-hour minimum engagement logic for hourly transport employees.
- **ATO (AU):** Tax Invoice compliance standards for sales $\geq \$1,000$.

2.5 Jurisdiction

This Agreement is governed by the laws of **Victoria, Australia**.

Signed by Client (Mehulkumar Patel, Director): _____ **Date:** _____

Signed by Developer (Authorized Representative): _____ **Date:** _____
